

CISD43_Final_Project

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0.1 USA Housing Dataset Analysis using Linear Regression

[2]:

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Course: CISD 43 – BIG DATA (Spring, 2026)

0.1.1 Problem Statement

- Using linear regression model, I want to find out what factors can accurately predict housing prices.
- **Keywords:** House price prediction, real estate , linear regression, data exploration,

0.1.2 Required packages

- Add instructions to install the required packages

```
[165]: ## Your code begins here
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
import numpy as np

%matplotlib inline
import warnings
warnings.filterwarnings('ignore')
```

0.1.3 Methodology

1. Using various libraries and linear regression to explore and predict housing prices
 - Steps:
 - Data loading and exploration
 - Data visualization
 - Linear Regression
2. Introduce the topics you used in your project

- Model 1
 - Linear Regression

0.1.4 Your code starts here

```
[160]: # load the dataset into a pandas Dataframe
# and displaying the first 5 rows of the dataframe
df = pd.read_csv('USA_Housing.csv')
df.head()
```

```
[160]: Avg. Area Income Avg. Area House Age Avg. Area Number of Rooms \
0      79545.458574      5.682861      7.009188
1      79248.642455      6.002900      6.730821
2      61287.067179      5.865890      8.512727
3      63345.240046      7.188236      5.586729
4      59982.197226      5.040555      7.839388

Avg. Area Number of Bedrooms Area Population Price \
0              4.09      23086.800503  1.059034e+06
1              3.09      40173.072174  1.505891e+06
2              5.13      36882.159400  1.058988e+06
3              3.26      34310.242831  1.260617e+06
4              4.23      26354.109472  6.309435e+05

Address
0  208 Michael Ferry Apt. 674\nLaurabury, NE 3701...
1  188 Johnson Views Suite 079\nLake Kathleen, CA...
2  9127 Elizabeth Stravenue\nDanielstown, WI 06482...
3              USS Barnett\nFPO AP 44820
4              USNS Raymond\nFPO AE 09386
```

```
[161]: # generating a quick statistical summary of a DataFrame's columns,
# helping you understand data distribution and spot anomalies at a glance
df.describe()
```

```
[161]: Avg. Area Income Avg. Area House Age Avg. Area Number of Rooms \
count      5000.000000      5000.000000      5000.000000
mean      68583.108984      5.977222      6.987792
std      10657.991214      0.991456      1.005833
min      17796.631190      2.644304      3.236194
25%      61480.562388      5.322283      6.299250
50%      68804.286404      5.970429      7.002902
75%      75783.338666      6.650808      7.665871
max      107701.748378      9.519088      10.759588

Avg. Area Number of Bedrooms Area Population Price
count      5000.000000      5000.000000  5.000000e+03
mean              3.981330      36163.516039  1.232073e+06
```

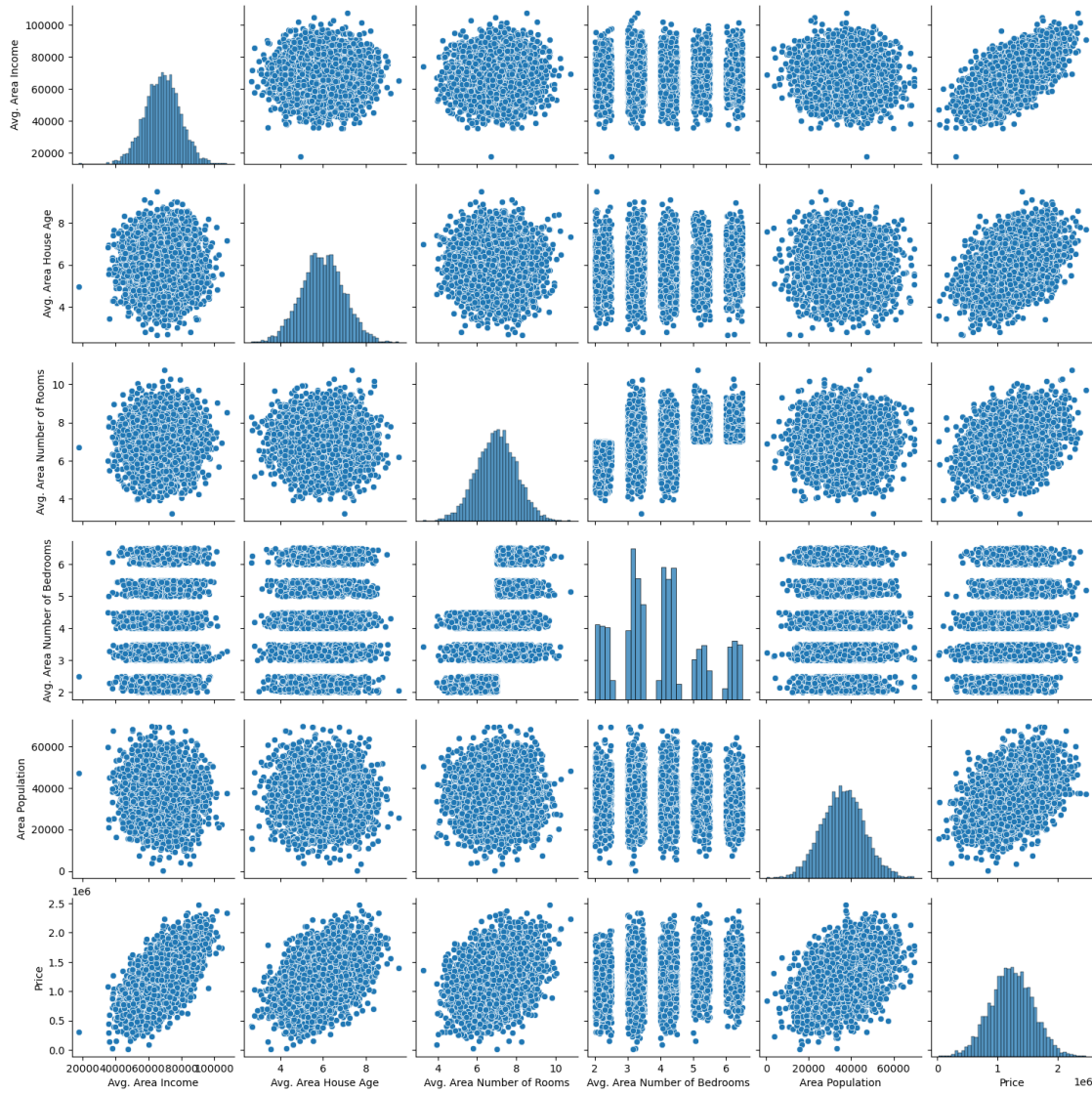
std	1.234137	9925.650114	3.531176e+05
min	2.000000	172.610686	1.593866e+04
25%	3.140000	29403.928702	9.975771e+05
50%	4.050000	36199.406689	1.232669e+06
75%	4.490000	42861.290769	1.471210e+06
max	6.500000	69621.713378	2.469066e+06

```
[162]: # generating a concise summary of the dataframe
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 7 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Avg. Area Income                      5000 non-null   float64
1   Avg. Area House Age                   5000 non-null   float64
2   Avg. Area Number of Rooms             5000 non-null   float64
3   Avg. Area Number of Bedrooms          5000 non-null   float64
4   Area Population                       5000 non-null   float64
5   Price                                 5000 non-null   float64
6   Address                               5000 non-null   object
dtypes: float64(6), object(1)
memory usage: 273.6+ KB
```

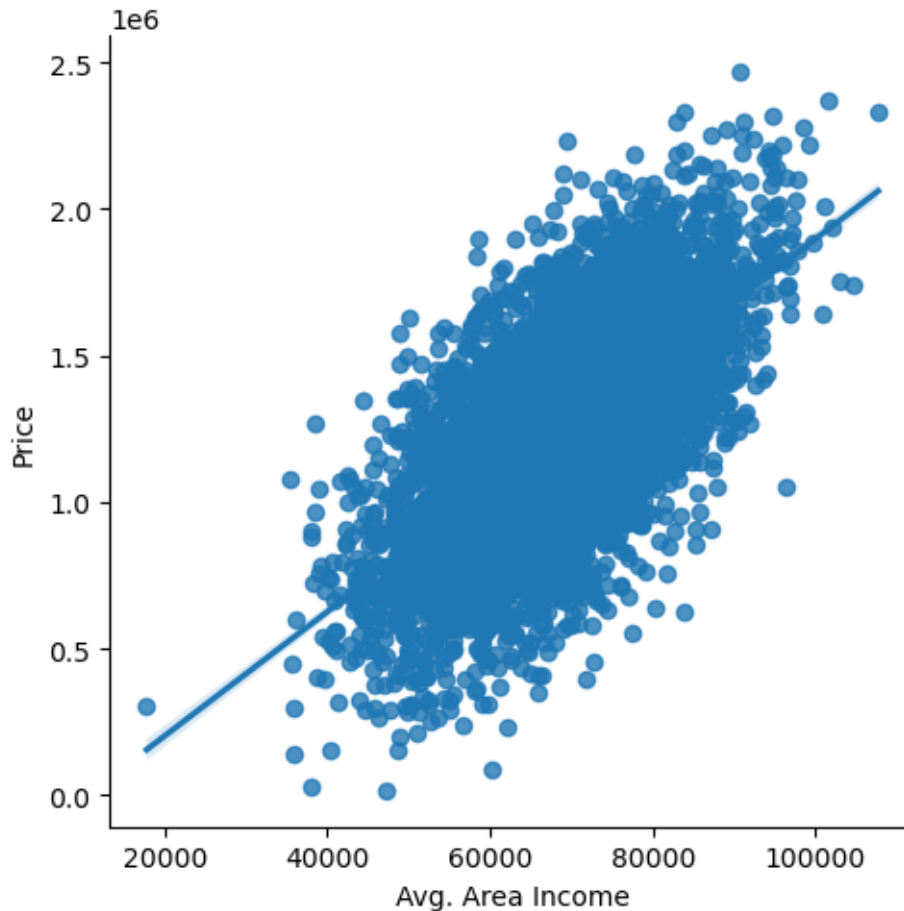
```
[163]: sns.pairplot(df)
```

```
[163]: <seaborn.axisgrid.PairGrid at 0x16ce50640>
```



```
[159]: sns.lmplot(x = 'Avg. Area Income', y = 'Price', data = df)
```

```
[159]: <seaborn.axisgrid.FacetGrid at 0x31a284550>
```



```
[153]: # generating the columns of the dataframe
df.columns
```

```
[153]: Index(['Avg. Area Income', 'Avg. Area House Age', 'Avg. Area Number of Rooms',
        'Avg. Area Number of Bedrooms', 'Area Population', 'Price', 'Address'],
        dtype='object')
```

```
[ ]:
```

```
[154]: X = df[['Avg. Area Income', 'Avg. Area House Age', 'Avg. Area Number of Rooms',
        'Avg. Area Number of Bedrooms', 'Area Population']]
y = df['Price']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.3,
        random_state = 42)

model = LinearRegression()
```

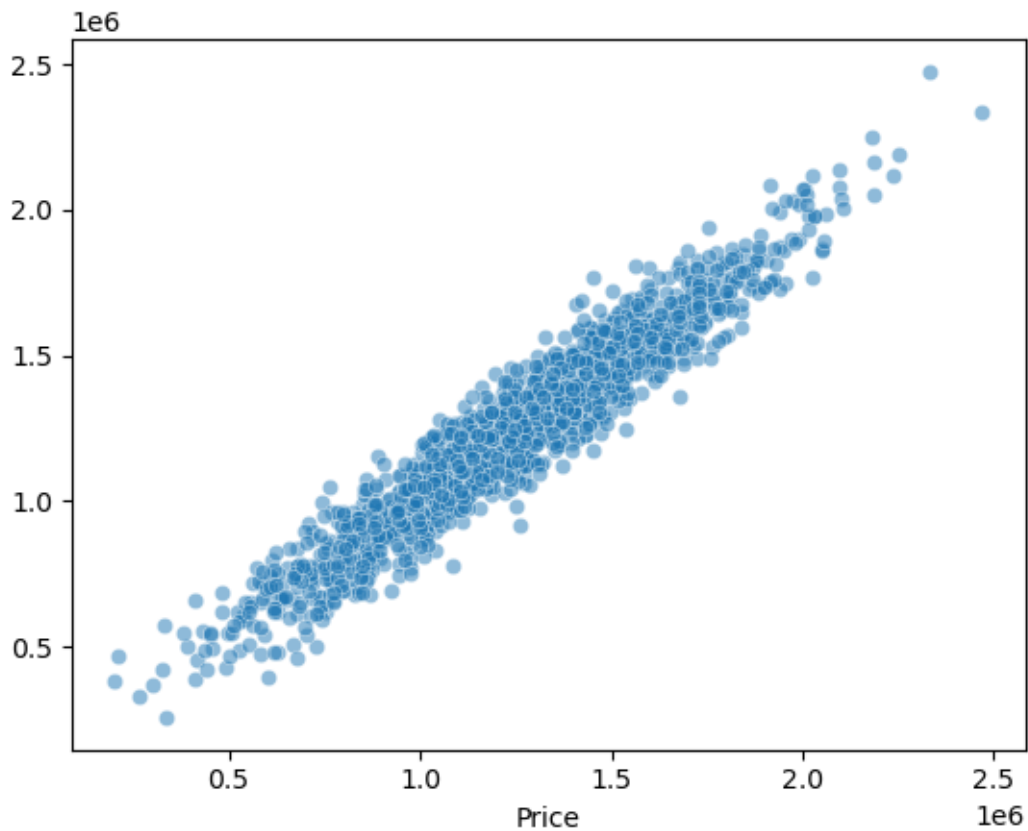
```
model.fit(X_train, y_train)

print(f'Coefficient: {model.coef_}')
```

```
Coefficient: [2.16257989e+01 1.65590393e+05 1.19827783e+05 2.36109526e+03
1.52165814e+01]
```

```
[155]: predictions = model.predict(X_test)
sns.scatterplot(x = y_test, y = predictions, alpha = 0.5)
```

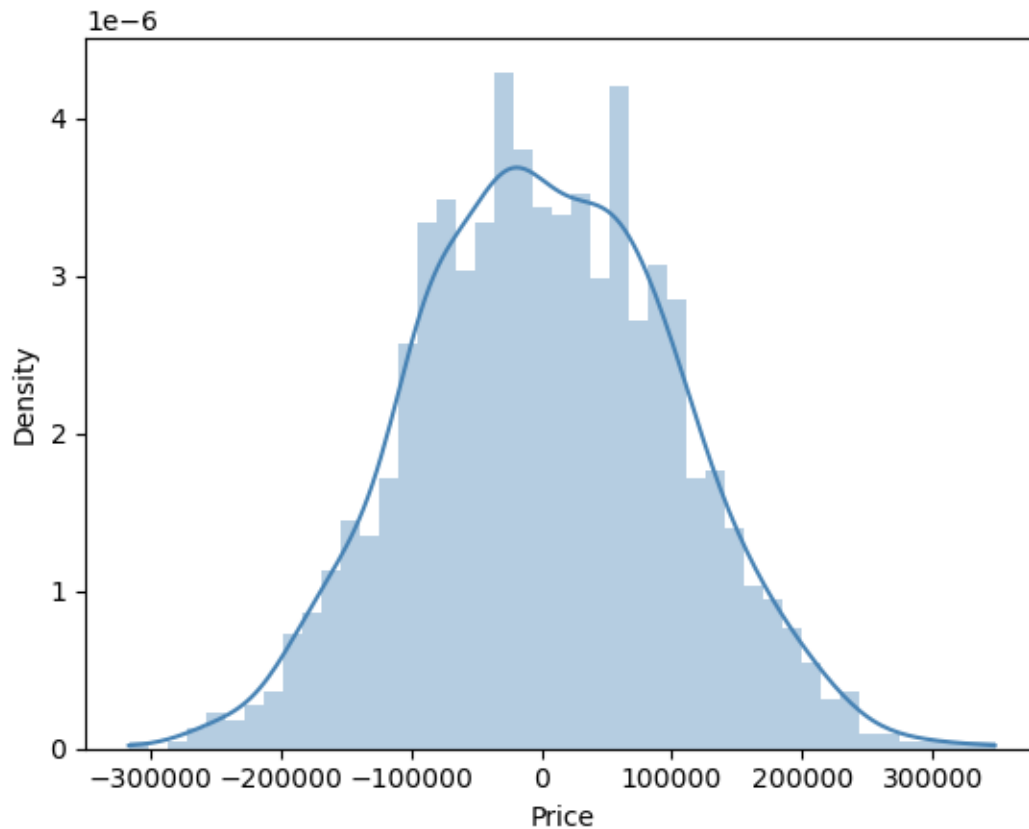
```
[155]: <Axes: xlabel='Price'>
```



```
[157]: residuals = y_test - predictions
sns.histplot(
    residuals,
    bins=45,
    kde=True,
    stat="density"
    ,
    color="steelblue"
```

```
,
alpha=0.4,
linewidth=0
)
```

[157]: <Axes: xlabel='Price', ylabel='Density'>



```
[158]: coefficients = pd.DataFrame(model.coef_,X.columns)
coefficients.columns = ['Coefficient']
coefficients
```

```
[158]:
```

	Coefficient
Avg. Area Income	21.625799
Avg. Area House Age	165590.392746
Avg. Area Number of Rooms	119827.783390
Avg. Area Number of Bedrooms	2361.095262
Area Population	15.216581

0.1.5 Conclusions

To answer the question of what factors contribute the housing prices the most, the model's coefficient indicates the age of the house and number of bedrooms are the biggest factors determining housing prices.

Model Reliability: The scatter plot of actual versus predicted prices demonstrates a strong linear alignment, confirming the model's accuracy across different price tiers. Furthermore, the normally distributed residuals indicate that the model's errors are random and unbiased, satisfying the core assumptions of linear regression.

[]: *# End of Project*